Homework Assignment 1
"Who ordered that?"

All problems are graded on effort only.

Griffiths: P1.1, P1.3, P1.7, P1.8

Problem 1: In lecture we discussed conservation of baryon number (A), strangeness (S), muon number (L_μ), electron number (L_e), charge (q), and energy (E). For each process below, identify one of the quantities above that is not conserved. Note that there are particle listings in the front matter of Griffiths.

(A) $\pi^+ + p^+ \rightarrow K^+ + \Sigma^0$

(B) $\pi^- + p^+ \rightarrow \Lambda + \Sigma^0$

(C) $\pi^- \rightarrow \mu^- + \nu_\mu$

(D) $p^+ \rightarrow \Delta^{++} + \pi^-$

(E) $\mu^- \rightarrow e^- + \nu_\mu$

(F) $K^0 \rightarrow \pi^+ + \pi^-$

Problem 2:

(A) Which of the above processes is allowed as a weak process?

(B) Transform one of the above processes to an allowed process using crossing symmetry and detailed balance.